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EDITORIAL

Who Liberates from the Ventilator and Who Decannulates from Tracheostomy? Insights from a Cervical Spinal Cord Injury Rehabilitation Cohort

Ellen Sloan,[1] Won Lee,[2] and Bethany L. Lussier[2,3,*]

Successful liberation from mechanical ventilation in cervical spinal cord injury (CSCI) requires an initial assessment of respiratory drive, ventilatory function, and control of impediments related to autonomic or central nervous system dysfunction accompanying the initial injury, followed by a systematic approach to liberation with standardized assessment of progress. The annual incidence of CSCI varies from 13 to 163.4 per million people worldwide, with mortality rates from 1.4 to 20%.[1] Such large regional variation in population and practice, variability in acute care outcomes, limited long-term survivor data, and paucity in access to specialist care available to survivors impede comprehensive research efforts. An estimated 90% of traumatic CSCI require invasive ventilation, with the majority undergoing tracheostomy, and ∼40% of those with CSCI remain ventilator dependent.[2,3] Life expectancy, healthcare costs, and quality of life have been tied to ventilator dependence,[4–6] prompting investigation into liberation strategies. The most successful weaning method to date has been periodic ventilator-free breathing accompanied by a combination of airway clearance and lung volume recruitment regimens.[7]

In the current issue of RESPIRATORY CARE, Fenger et al provide a retrospective analysis of a cohort of CSCI patients who were admitted to a specialized rehabilitation hospital where they underwent protocolized multimodal respiratory therapy and standardized ventilator liberation using a progressive ventilator-free breathing model. In this study, 39 subjects underwent progressive ventilator-free breathing according to protocol, with 62% being liberated from invasive ventilatory support, 72% weaned to nocturnal support only, and 48% fully decannulated at the time of discharge. Though a modest sample size, this cohort of subjects with CSCI who survived acute ICU admission to discharge to specialized spinal cord injury weaning facility is among the largest in the literature and provides important information regarding utility of protocolized ventilator liberation and overall prognosis.[8]

In the rehabilitation phase, Fenger and colleagues detail a standardized respiratory care approach with 8–12 mL/kg tidal volume ventilation and augmented airway clearance.[8] There are limited antiquated data (and persistent controversy) that support the safety of very high tidal (10–15 mL/kg) volume targets compared with low (rather than medium) tidal volumes, but the superiority of this method when standard ventilation augmented by atelectasis prevention/lung volume recruitment or in transitional attempts to wean remains highly debated.[9,10] Airway clearance, breath-stacking, and hyperinflation are considered part of standard respiratory care and presumably augmented the ventilatory approach in this cohort without targeting very high tidal volumes.[7,11,12] Existing literature reports failure to wean from ventilatory support was associated with age >56, vital capacity <1,500 mL, obesity with body mass index >25, or presence of underlying obstructive pulmonary disease.[13,14] Implementation of structured respiratory rehabilitation protocols demonstrated success of 70–92% even in difficult to wean subjects previously discharged home requiring mechanical ventilation.[10,12]

In keeping with prior literature, Fenger and colleagues demonstrate that subjects with lower level CSCI (C5–C8) were more likely to achieve full liberation from ventilation and decannulation compared with higher CSCI (C1–C4)

[1]Dr. Sloan is affiliated with Division of Physical Medicine and Rehabilitation, UT Southwestern Medical Center, Dallas, Texas, USA.
[2]Dr. Lee and Lussier are affiliated with Division of Pulmonary Critical Care, UT Southwestern Medical Center, Dallas, Texas, USA.
[3]Dr. Lussier is affiliated with Division of Neurological Critical Care, UT Southwestern Medical Center, Dallas, Texas, USA.

*Address Correspondence to: Bethany L Lussier, MD, Pulmonary and Critical Care Medicine, UT Southwestern Medical Center, 5323 Harry Hines Blvd, Dallas, Texas 75390, Email: Bethany.Lussier@utsouthwestern.edu

with an OR of 5.54. These findings are difficult to compare with prior studies where liberation and decannulation rates vary significantly, from 45 to 85% and 25 to 100%, respectively.[15–17] In this study, median stay at a specialized rehabilitation center was 64 days (IQR 56–76), and of those, 74% (29/39) were weaned from daytime ventilatory support by the time of discharge. Of the 39 subjects discharged at the time, 59% (n = 23) were completely liberated from invasive ventilation, and 46% (n = 18) were fully decannulated at a median day of 36.[8] In this study by Fenger et al, low CSCI was with a 5.54 (1.51–20.4, P = .01) higher odds of ventilatory independence at discharge. Interestingly, low CSCI was not independently associated with discharge level of care, but older age was associated with discharge to more resource-intensive settings. As expected, even after adjusting for age, decannulation was significantly associated with home discharge (proportional OR 22.84 [2.84–183.57, P = .003]) compared with those unable to wean.

It is important to consider that patients who undergo intensive weaning programs, although in the ICU prior to inpatient rehabilitation for extended stay, may have different outcomes to those who undergo weaning protocols at a specialist weaning spinal cord injury center, non-weaning spinal cord injury center, or non-spinal cord injury rehab.[10] Successful liberation in a recent study published in 2025, in fact, demonstrated that 78% of subjects with high CSCI (C2–C4) (n = 18) undergoing multimodal respiratory weaning interventions were liberated from mechanical ventilation, but discharge to home was only in 7% of the entire cohort, and the majority of the intervention took place during the acute ICU admission.[10] This is quite important to consider in the context of the current research by Fenger et al, as patients discharged to specialty rehabilitation centers have already been determined to be challenging to wean and have may have had an extended stay in an ICU prior to a spinal cord injury rehabilitation encounter. The reason this comparison is important is that the cohort discussed in the current issue of Respiratory Care by Fenger and colleagues are patients who survived to ICU discharge and remained ventilator dependent at a median day 17 (13–25) post injury, and interventions studied in the current issue reflect independent practice subsequent to ICU care at an independent ventilator weaning rehabilitation facility. This important fact must be considered when attempting to use the findings to predict outcomes in similar populations.

Another strength of this investigation that is unfortunately lacking in so many studies was the long-term follow-up with a median of 1,101 days for the cohort. Though not a primary outcome of this study, it is notable that of the 26% of subjects who died at a median of 382 days, discharge location rather than ventilation status predicted time to mortality.

The limitations of a retrospective analysis need to be considered when applying the information gathered here to a large heterogenous CSCI population, particularly if proposing use of these data in supporting prognostication. The retrospective nature cannot control for the potential impact of confounding variables that could affect liberation and decannulation rates, such as consistent adherence to airway clearance strategies, aspiration pneumonia rates, and qualitative assessment of bulbar function. The impact of favorable physical, occupational, and speech therapy on ventilator liberation is also difficult to measure, as even standardized protocols for intervention are personalized based on response to intervention and progress.

The nature of a retrospective study limits the assessment of impact of autonomic dysfunction in the CSCI population, as typically immediately addressed but can influence the tolerance of certain respiratory clearance strategies. For example, unopposed parasympathetic activity in CSCI and increased secretions, hemodynamic variability in response to hyperinflation therapy in terms of hemodynamics, and response to intervention are rarely documented in a systematic way. Risk of depression and anxiety limiting engagement and preexisting medical and psychiatric conditions may be associated with higher risk of breathing independence failure in older individuals. Although not mentioned in the article, CSCI is commonly encountered with comorbid traumatic brain injury that may influence level of engagement, attention, and ability to follow commands and may have variable effect on implementation of protocolized therapy.

In future studies it would be interesting if respiratory weaning and tracheostomy decannulation protocols could be personalized based on initial versus summative CSCI level and impact on respiratory mechanics. For example, studying the potential for engagement of diaphragm muscle strengthening and if phrenic nerve stimulation could impact liberation outcomes in patients with ability to trigger inspiratory breaths. Patients who can maintain brief periods of self-ventilation would be eligible to participate in ventilator-free breathing weaning protocols. Previously, stimulation of the phrenic nerve and diaphragm was reserved for patients with near-complete inability to trigger diaphragm function because of high-level injury with permanent ventilatory support needs (C1–C3 injury).[11,18] In more recent years there has been emerging innovative research into stimulation as a potential means to expedite or augment phrenic nerve and diaphragm recovery for lower levels of cervical spinal cord or nerve injury. How nerve stimulation might impact recovery of diaphragm or phrenic nerve function and potential for decannulation in a spinal cord injury ventilator rehabilitation program,

particularly in the lower CSCI (C5–C7) population, will be of future interest.

This article by Fenger and colleagues highlights that a systematic, protocolized approach to ventilator management, patience, and consistency are key to successful mechanical ventilation in CSCI. After admission to rehabilitation, the timeline for ventilator liberation and decannulation rates are informative for the healthcare team in terms of resource allocation and outcome expectation concordance. Although formal prognostication of breathing independence cannot be generalized based on a conservative sample size from an experienced single center, this research meaningfully adds to the growing literature to support best efforts in respiratory care following CSCI.

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